

SIBGATUL HASSEN

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PROFESSIONAL SUMMARY

Motivated Computer Science & Engineering undergraduate student with a strong background in software engineering, machine learning research, quantum-classical hybrid architectures, bioinformatics, computer networking, and full-stack web development. Proven track record in rapid prototyping during technical hackathons and academic research projects. Seeking an entry-level software engineering or technical research internship role to apply problem-solving skills, concurrent programming knowledge, and quality assurance principles.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, PHP, TypeScript, SQL, Bash / Shell Scripting, PowerShell
- **AI, ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, River, Hugging Face, LoRA, Large Language Models (LLM), Image Processing, Quantum ML (PennyLane), TextBlob, Transformers (RoBERTa), Model Evaluation, Data Preprocessing
- **Bioinformatics & Computational Biology:** Protein Analysis, Biological Sequence Analysis, Antimicrobial Peptide Generation, Machine Learning for Biological Applications, AlphaFold
- **Web & Databases:** React, Node.js, Express.js, HTML5/CSS3, RESTful APIs, PostgreSQL, MongoDB, MySQL, Prisma, DBMS
- **Networking & Embedded:** Wireshark, Cisco Packet Tracer, CupCarbon (WSN Simulation), Assembly (8086), Microprocessor Systems, Arduino, Sensors (Ultrasonic, MPU6050, LDR), Electronics, IoT
- **Tools, Platforms & Design:** Kaggle, Google Colab, Git/GitHub, GitBash, Linux/WSL, Figma, LaTeX, MS Office, Agile Methodology, Computer Security Principles, SQA & Unit Testing

EDUCATION

Bachelor of Science in Computer Science & Engineering

United International University (UIU) | Dhaka, Bangladesh

Expected Graduation: February 2027

Final Year Design Project (GITHUB) : Disulfide-Topology-Conditioned Generation of Defensin-Like Antimicrobial Peptides

RESEARCH PROJECTS

QMedLeafNet – Quantum-Classical Hybrid Model for Plant Health Detection (*Digital Image Processing Research* | Submitted to ICCIT 2026 | MobileNetV3, Quantum ML) [GITHUB](#)

- Developed a plant disease classification system using a MobileNetV3-Small backbone and quantum-classical hybrid architectures.
- Evaluated models on 25,962 images across 63 classes using five-fold cross-validation.
- Compared quantum-inspired and classical classification heads using parameter-matched experiments.
- Manuscript submitted to ICCIT 2026.

MechanismSelector – Energy-Efficient Concept Drift Adaptation (*Ongoing Green Computing Research | Scikit-learn, XGBoost, River, Python*) [GITHUB](#)

- Investigating adaptive retraining strategies for machine learning models under concept drift.
- Comparing model nudging and complete retraining based on computational cost and performance.

Privacy Policy Comprehension (*Ongoing HCI Research | Human-Computer Interaction*) [GITHUB](#)

- Contributing to a Bangla-English study design comparing privacy-policy comprehension, user confidence, and reading time among university students.

Bank Marketing Data Mining & Bangla LLM Explainer (*Ongoing Data Mining Research | Python, Scikit-learn, Tabular ML, LLMs*) [GITHUB](#)

- Performing data mining on the UCI Bank Marketing dataset incorporating modern tabular machine learning techniques.
- Building visual workflow pages and developing an integrated Bangla LLM explainer for model interpretation.

PROJECTS

LabLink – Lab Management Platform (*Software Engineering Lab Project | Node.js, Express, DBMS*) [GITHUB](#)

- Developed a full-featured web application for booking lab equipment, tracking consumables, and logging breakdown reports for CSE lab courses.
- Designed database schemas and implemented user authentication workflows to streamline lab administration.

UIU Talent Hunt – Full-Stack Talent Discovery Platform (*Web Programming Project | React, TypeScript, Node.js, MongoDB*) [GITHUB](#)

- Built a full-stack platform with user authentication, audio content management, player interfaces, and administrative functionality.
- Developed the frontend using React and TypeScript, with a Node.js backend and MongoDB database.

Dhaka/Bangladesh Reddit Data & Sentiment Analysis Pipeline (*ML Project | Python, Pandas, TextBlob, RoBERTa/Transformers*) [GITHUB](#)

- Engineered an end-to-end data collection, deduplication, and sentiment/emotion analysis pipeline over 1,854 unique Reddit posts from r/dhaka and r/bangladesh.
- Built custom scripts for keyword and area-based crawling, data cleaning, subjectivity/polarity scoring, and topic categorization.
- Developed automated report generators producing interactive HTML dashboards and Markdown summaries for community discussion analysis.

Computer Networking & Simulation Projects (*Academic Networking Projects | Cisco Packet Tracer, Wireshark, CupCarbon*)

- Designed and configured network topologies, subnetting, and routing protocols using Cisco Packet Tracer.
- Analyzed network traffic, packet captures, and protocol headers using Wireshark.
- Simulated Wireless Sensor Networks (WSN) and IoT environments using CupCarbon.

Personalized Meal Planner (*DBMS Project | Database Systems, Web Tech*) [GITHUB](#)

- Designed a database-driven application tracking user health profiles and preferences to suggest recipes and meal plans.
- Integrated vendor database connectivity for automated ordering of ingredients required for selected recipes.

Mental Asylum & Rehabilitation Center Management System (*Computer Security Project*) [GITHUB](#)

- Developed a management system for a mental health and rehabilitation center.
- Designed secure system workflows for organizing patient, staff, and rehabilitation-related information.

Video Game Vault (*Advanced Object Oriented Programming Project | Full-Stack Web Tech*) [GITHUB](#)

- Built a full-stack web application designed to catalog video games with interactive status tracking, cover art rendering, and technical documentation.

- Applied software engineering principles to structure the system and manage its core functionality.

Microprocessor Systems & Assembly Implementation (*Academic Lab Project (CSE-4326) | Assembly Language, 8086 Microprocessor*)

[GITHUB](#)

- Executed hardware-level microprocessor experiments and low-level assembly programming for the CSE-4326 course at UIU.

Gym Recommendation System (*Artificial Intelligence Project | Python, Machine Learning*)

[GITHUB](#)

- Designed and implemented an algorithmic recommendation engine suggesting tailored workout routines and fitness plans based on user profiles.

Smart Parking Unit Testing Suite (*Software Quality Assurance Project | Python, Testing Frameworks*)

[GITHUB](#)

- Built structured SQA test plans and automated unit test cases to evaluate core system functionalities and edge cases of a smart parking system.

CATWIZARD – Terminal Customization Utility (*Personal Project | Bash Scripting, Linux / WSL*)

[GITHUB](#)

- Engineered a lightweight custom Linux CLI script inspired by `cowsay` featuring customized ASCII art output.

HACKATHONS & EXTRACURRICULAR ACTIVITIES

- **BUET CSE FEST Hackathon 2026** – Team Member | Rapid Prototyping & System Design
- **BUP Hackathon 2026** – Participant | Collaborative Backend Development
- **BCF Hackathon Preliminary 2026** – Team Member | Software Solutions Development
- **UIU Hackathon** – Participant | Project Prototyping & Technical Solution Design

RELEVANT COURSEWORK

Data Structures & Algorithms • Object-Oriented Programming (OOP) • Database Management Systems (DBMS) • Software Engineering • Artificial Intelligence • Computer Networks & Security • Microprocessors & Embedded Systems • Operating Systems • Concurrent Programming • Computational Biology • Machine Learning • Data Mining • Digital Image Processing • Human Computer Interaction • Software Testing & Quality Assurance

ADDITIONAL INFORMATION

- Antimicrobial Peptide Generation • Protein Analysis • Machine Learning for Biological Applications • Bioinformatics • Efficient Model Adaptation • Large Language Models (LLMs) • Image Processing
- **Portfolio:** <https://sibgatul-hassen.github.io/Portfolio/>